

Applied Chemistry

SITAMS CSE (AI & ML) • R23 • I Year • I Semester • Course Code: 23BSC111T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: STRUCTURE AND BONDING MODELS

Syllabus topics

- Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2 , particle in one dimensional box, molecular orbital theory – bonding in homo- and heteronuclear diatomic molecules – energy level diagrams of O_2 and CO , etc
- π - molecular orbitals of butadiene and benzene, calculation of bond order

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Source page(s): 34

Unit II: MODERN ENGINEERING MATERIALS

Syllabus topics

- Semiconductors, band diagram in solids, Semiconductor devices (p-n junction diode as rectifier and transistors) Superconductors-Introduction basic concept, applications
- Supercapacitors: Introduction, Basic Concept-Classification – Applications
- Nano materials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphines nanoparticles

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Unit III: ELECTROCHEMISTRY AND APPLICATIONS

Syllabus topics

- Electrochemical cell, Nernst equation, cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), concept of conductivity, conductivity cell, conductometric titrations (acid-base titrations)
- Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples
- Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries- working of the batteries including cell reactions
- Fuel cells, hydrogen-oxygen fuel cell– working of the cells
- Polymer Electrolyte Membrane Fuel cells (PEMFC)

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Unit IV: POLYMER CHEMISTRY

Syllabus topics

- Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation
- Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of – PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres
- Elastomers–Buna-S, Buna-N–preparation, properties and applications
- Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications
- Bio-Degradable polymers - Poly Glycolic Acid (PGA), Polyl Lactic Acid (PLA)

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Unit V: INSTRUMENTAL METHODS AND APPLICATIONS

Syllabus topics

- Electromagnetic spectrum
- Absorption of radiation: Beer-Lambert's law
- UV-Visible Spectroscopy, electronic transition, Instrumentation, IR spectroscopies, fundamental modes and selection rules, Instrumentation
- Chromatography-Basic Principle, Classification-HPLC: Principle, Instrumentation and Applications

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This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.